HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **query optimization strategies** for a distributed database engine, reducing average query latency by **28%**.
- Developed **new features** for the MongoDB query language parser and optimizer, enhancing support for complex analytical workloads.
- Contributed to the **geospatial search subsystem**, improving indexing performance for location-based queries by **18%**.
- Refactored **large code bases** in C++ within the query execution layer, improving maintainability and reducing bug introduction rates.
- Collaborated with **client drivers and cloud services teams** to ensure seamless integration of query engine updates into downstream products.
- Participated in **code reviews** for new query engine features and design proposals, ensuring adherence to architectural standards and best practices.
- Containerized **query engine components** using Docker and deployed them to a Kubernetes cluster for efficient testing and development workflows.

SKILLS

Database Internals – Query Optimization, Query Language, Query Execution, Parsing, Plan Selection, Indexing, Geospatial Search, Distributed Systems

Programming Languages – C++, Java, Python, SQL

Databases – MongoDB, PostgreSQL, MySQL, Oracle

Cloud & Orchestration – AWS, Docker, Kubernetes

Development Practices – Algorithms, Data Structures, Software Design, Agile Methodologies, Code Review, CI/CD

Analytics & Data – Data Warehousing, ETL, Power BI, Tableau, R, scikit-learn

PROJECTS

Distributed Query Engine Performance Enhancements (HCL Technologies)

- Optimized **SQL query execution plans** within an enterprise data platform, resulting in a **34% reduction** in report generation times for key business intelligence dashboards.
- Developed Python scripts to automate the extraction and transformation of data from **PostgreSQL and Oracle** databases, feeding into scheduled operational reports.
- Implemented robust error handling and logging for data ingestion pipelines using **Java Spring Boot**, improving system stability during peak processing hours.

Database System Design & Analysis (MSc Applied Project)

- Designed and implemented a **relational database schema** for analytical workloads, incorporating indexing strategies that improved query performance by **22%** on a 500GB dataset.
- Built interactive **Power BI dashboards** to visualize query performance metrics and identify bottlenecks, providing actionable insights to a team of 6 data engineers.
- Analyzed query patterns using **SQL** and conducted statistical analysis in R to understand user behavior and inform future database design decisions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)